1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 12:12 AM
State	Finished
Completed on	Friday, 9 October 2026, 12:16 AM

Started on	Friday, 9 October 2026, 12:12 AM
State	Finished
Completed on	Friday, 9 October 2026, 12:16 AM

Time taken	3 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
```

```
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                printf("%d",arr[i]);
                return 0;
            }
        }
    }
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 12:16 AM
State	Finished
Completed on	Friday, 9 October 2026, 12:48 AM
Time taken	31 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    int hash[n+1];
    for(int i=0;i<n;i++){
        hash[i]=0;
    }
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++){
        if(hash[arr[i]]==1){
            printf("%d",arr[i]);
        }
        hash[arr[i]]=1;
    }
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 12:48 AM
State	Finished
Completed on	Friday, 9 October 2026, 1:13 AM
Time taken	24 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int T,n1,n2;
    scanf("%d",&T);
    while(T--){
    scanf("%d",&n1);
    int arr[n1];
    for(int i=0;i<n1;i++){
        scanf("%d",&arr[i]);
    }
    scanf("%d",&n2);
    int arr2[n2];
    for(int i=0;i<n2;i++){
        scanf("%d",&arr2[i]);
    }
    for(int i=0;i<n1;i++){
        for(int j=0;j<n2;j++){
            if(arr[i]==arr2[j]){
                printf("%d ",arr[i]);
                break;
            }
            if(arr2[j]>arr[i]){
                break;
            }
        }
    }
    printf("\n");
}
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RAJALAKSHMI ENGINEERING COLLEGE

Malavika M 250701409-2025-CSE M2

Dashboard My courses

CS23331-DAA-2025-CSE / 4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 1:13 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:21 AM
Time taken	1 hour 7 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1
2  #include <stdio.h>
3
4  int main() {
5      int T;
6      scanf("%d", &T);
```

```c
int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int m, n;

        scanf("%d", &m);
        int a[m];

        for (int i = 0; i < m; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n);
        int b[n];

        for (int i = 0; i < n; i++) {
            scanf("%d", &b[i]);
        }

        int i = 0, j = 0;

        while (i < m && j < n) {
            if (a[i] == b[j]) {
                if(a[i]!=a[i-1]){
                    printf("%d ", a[i]);
                }

                int value = a[i];


                while (i < m && a[i] == value) {
                    i++;
                }
                while (j < n && b[j] == value) {
                    j++;
                }
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 2:22 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:30 AM
Time taken	8 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5	1

1 3 5 4	

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int k;
    scanf("%d",&k);
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[j]-a[i]==k){
                printf("1");
                return 0;
            }
        }
    }
    printf("0\n");
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 2:31 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:59 AM
Time taken	28 mins 5 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int k;
    scanf("%d",&k);
    int i=0,j=1;
    while(i<n&&j<n){
        if(i==j){
            j++;
            continue;
        }
        int diff=arr[j]-arr[i];
        if(diff==k){
            printf("1\n");
```

```c
    scanf("%d",&k);
    int i=0,j=1;
    while(i<n&&j<n){
        if(i==j){
            j++;
            continue;
        }
        int diff=arr[j]-arr[i];
        if(diff==k){
            printf("1\n");
            return 0;
        }
        else if(diff<k){
            j++;
        }
        else{
            i++;
        }

    }
    printf("0\n");
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.